Supplementary Information: On the Accuracy of Point Estimation Methods for Moment-Based Reliability and Sensitivity Analysis

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1 Error computed from RM for independent system of random variables.

$$
\begin{aligned}
\text{Error} \;=\; & \sum_{j=4}^{\infty}\frac{1}{j!}\sum_{i=1}^{n}\frac{\partial^{j} f(\boldsymbol{\mu})}{\partial \xi_i^{j}}\left\{\mathbb{E}(\xi_i-\mu_i)^{j}-\sigma_i^{j}\sum_{k_i=1}^{2}\omega_{i,k_i}(\eta_{k_i}\alpha_{i,k_i})^{j}\right\}\\
+\; & \sum_{j_1=2}^{\infty}\sum_{j_2=2}^{\infty}\frac{1}{j_1!j_2!}\sum_{i_1<i_2}\frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1}\partial \xi_{i_2}^{j_2}}\Bigg\{\mathbb{E}(\xi_{i_1}-\mu_{i_1})^{j_1}\mathbb{E}(\xi_{i_2}-\mu_{i_2})^{j_2}\\
-\; & \sigma_{i_1}^{j_1}\sigma_{i_2}^{j_2}\sum_{k_{i_1}=1}^{2}\omega_{i_1,k_{i_1}}(\eta_{k_{i_1}}\alpha_{i_1,k_{i_1}})^{j_1}\sum_{k_{i_2}=1}^{2}\omega_{i_2,k_{i_2}}(\eta_{k_{i_2}}\alpha_{i_2,k_{i_2}})^{j_2}\Bigg\}\\
+\; & \sum_{j_1=2}^{\infty}\sum_{j_2=2}^{\infty}\sum_{j_3=2}^{\infty}\frac{1}{j_1!j_2!j_3!}\sum_{i_1<i_2<i_3}\frac{\partial^{j_1+j_2+j_3} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1}\partial \xi_{i_2}^{j_2}\partial \xi_{i_3}^{j_3}}\Bigg\{\mathbb{E}(\xi_{i_1}-\mu_{i_1})^{j_1}\ldots\mathbb{E}(\xi_{i_3}-\mu_{i_3})^{j_3}\\
-\; & \sigma_{i_1}^{j_1}\sigma_{i_2}^{j_2}\sigma_{i_3}^{j_3}\sum_{k_{i_1}=1}^{2}\omega_{i_1,k_{i_1}}(\eta_{k_{i_1}}\alpha_{i_1,k_{i_1}})^{j_1}\sum_{k_{i_2}=1}^{2}\omega_{i_2,k_{i_2}}(\eta_{k_{i_2}}\alpha_{i_2,k_{i_2}})^{j_2}\sum_{k_{i_3}=1}^{2}\omega_{i_3,k_{i_3}}(\eta_{k_{i_3}}\alpha_{i_3,k_{i_3}})^{j_3}\Bigg\}\\
+\; & \sum_{j_1=2}^{\infty}\cdots\sum_{j_n=2}^{\infty}\frac{1}{j_1!\ldots j_n!}\sum_{i_1<\ldots<i_n}\frac{\partial^{j_1+\cdots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1}\ldots\partial \xi_{i_n}^{j_n}}\Bigg\{\mathbb{E}(\xi_{i_1}-\mu_{i_1})^{j_1}\ldots\mathbb{E}(\xi_{i_n}-\mu_{i_n})^{j_n}\\
-\; & \sigma_{i_1}^{j_1}\ldots\sigma_{i_n}^{j_n}\sum_{k_{i_1}=1}^{2}\omega_{i_1,k_{i_1}}(\eta_{k_{i_1}}\alpha_{i_1,k_{i_1}})^{j_1}\ldots\sum_{k_{i_n}=1}^{2}\omega_{i_n,k_{i_n}}(\eta_{k_{i_n}}\alpha_{i_n,k_{i_n}})^{j_n}\Bigg\}.
\end{aligned}
\tag{1.1}
$$

2 Error computed from HM for independent system of random variables.

$$
\begin{aligned}
\text{Error} \;=\; & \sum_{j=3}^{\infty}\frac{1}{j!}\sum_{i=1}^{n}\frac{\partial^{j} f(\boldsymbol{\mu})}{\partial \xi_i^{j}}\left\{\mathbb{E}(\xi_i-\mu_i)^{j}-\frac{1}{2}\sigma_i^{j}n^{j/2-1}(1+(-1))^{j}\right\}\\
+\; & \sum_{j_1=2}^{\infty}\sum_{j_2=2}^{\infty}\frac{1}{j_1!j_2!}\sum_{i_1<i_2}\frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1}\partial \xi_{i_2}^{j_2}}\,\mathbb{E}\left((\xi_{i_1}-\mu_{i_1})^{j_1}(\xi_{i_2}-\mu_{i_2})^{j_2}\right)+\cdots\\
+\; & \sum_{j_1=2}^{\infty}\cdots\sum_{j_n=2}^{\infty}\frac{1}{j_1!\ldots j_n!}\sum_{i_1<\ldots<i_n}\frac{\partial^{j_1+\cdots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1}\ldots\partial \xi_{i_n}^{j_n}}\,\mathbb{E}(\xi_{i_1}-\mu_{i_1})^{j_1}\ldots\mathbb{E}(\xi_{i_n}-\mu_{i_n})^{j_n}.
\end{aligned}
\tag{2.1}
$$

3 Error computed from LM for independent system of random variables.

$$
\begin{aligned}
\text{Error} \quad = \quad & \sum_{j=5}^{\infty} \frac{1}{j!} \sum_{i=1}^{n} \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \left\{ \mathbb{E}(\xi_i - \mu_i)^j - \sigma_i^j \left(\omega_i^+ (\alpha_i^+)^j + \omega_i^- (\alpha_i^-)^j \right) \right\} \\
+ \quad & \sum_{j_1=2}^{\infty} \sum_{j_2=2}^{\infty} \frac{1}{j_1! j_2!} \sum_{i_1<i_2} \frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2}} \, \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \mathbb{E}(\xi_{i_2} - \mu_{i_2})^{j_2} \\
+ \quad & \sum_{j_1=2}^{\infty} \cdots \sum_{j_n=2}^{\infty} \frac{1}{j_1! \ldots j_n!} \sum_{i_1<\ldots<i_n} \frac{\partial^{j_1+\cdots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \ldots \partial \xi_{i_n}^{j_n}} \, \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \ldots \mathbb{E}(\xi_{i_n} - \mu_{i_n})^{j_n} .
\end{aligned}
\tag{3.1}
$$

4 Error computed from MHM for independent system of random variables.

$$
\begin{aligned}
\text{Error} \quad = \quad & \sum_{j=3}^{\infty} \frac{1}{j!} \sum_{i=1}^{n} \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \left\{ \mathbb{E}(\xi_i - \mu_i)^j - \frac{1}{2} \sigma_i^j n^{j/2-1} (1 + (-1))^j \right\} \\
+ \quad & \sum_{j_1=2}^{\infty} \sum_{j_2=2}^{\infty} \frac{1}{j_1! j_2!} \sum_{i_1<i_2} \frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2}} \, \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \mathbb{E}(\xi_{i_2} - \mu_{i_2})^{j_2} + \cdots \\
+ \quad & \sum_{j_1=2}^{\infty} \cdots \sum_{j_n=2}^{\infty} \frac{1}{j_1! \ldots j_n!} \sum_{i_1<\ldots<i_n} \frac{\partial^{j_1+\cdots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \ldots \partial \xi_{i_n}^{j_n}} \, \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \ldots \mathbb{E}(\xi_{i_n} - \mu_{i_n})^{j_n} .
\end{aligned}
\tag{4.1}
$$

5 Response mean approximated from DRM

Substituting $s = 1$ in Eqn (2.6.1) of the main text and applying expectation operator on its both sides, we get the approximated response mean as:

$$
\widehat{\mu}_Y = \sum_{i=1}^{n} \mathbb{E} \left(f(\mu_1, \ldots, \mu_{i-1}, \xi_i, \mu_{i+1}, \ldots, \mu_n) \right) - (n-1) f(\boldsymbol{\mu}) \, . \tag{5.1}
$$

Now v number of concentration points $\xi_{i,k} = \mu_i + \alpha_{i,k}\sigma_i$ with the corresponding weights $\omega_{i,k}$, $k = 1, 2, \ldots, v$ are selected for each random variable ξ_i, $i = 1, 2, \ldots, n$. The term $f(\mu_1, \ldots, \mu_{i-1}, \xi_{i,k}, \mu_{i+1}, \ldots, \mu_n)$ is expanded in Taylor's series about the input mean $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_n)$. This yields

$$
\widehat{\mu}_Y = \sum_{i=1}^{n} \sum_{k=1}^{v} \omega_{i,k} \left(f(\boldsymbol{\mu}) + \sum_{j=1}^{\infty} \frac{1}{j!} \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} (\alpha_{i,k} \sigma_i)^j \right) - (n-1) f(\boldsymbol{\mu}) \, . \tag{5.2}
$$

It may be noted that $\sum_{k=1}^{v} \omega_{i,k} = 1$, then

$$
\widehat{\mu}_Y = f(\boldsymbol{\mu}) + \sum_{j=1}^{\infty} \frac{1}{j!} \sum_{i=1}^{n} \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \sigma_i^j \sum_{k=1}^{v} \omega_{i,k} \alpha_{i,k}^j \, . \tag{5.3}
$$

Now, let us consider $s = 2$. The estimated mean ($\widehat{\mu}_Y$) of the response function is given by:

$$
\begin{aligned}
\widehat{\mu}_Y \quad = \quad & \sum_{i_1<i_2} \mathbb{E}(f(\mu_1, \ldots, \mu_{i_1-1}, \xi_{i_1}, \mu_{i_1+1}, \ldots, \mu_{i_2-1}, \xi_{i_2}, \mu_{i_2+1}, \ldots, \mu_n)) \\
- \quad & (n-2) \sum_{i=1}^{n} \mathbb{E}(f(\mu_1, \ldots, \mu_{i-1}, \xi_i, \mu_{i+1}, \ldots, \mu_n)) + \frac{(n-1)(n-2)}{2} f(\boldsymbol{\mu}) \, .
\end{aligned}
\tag{5.4}
$$

The expectation terms are obtained using the product rule of numerical integration as follows:

$$\widehat{\mu}_Y = \sum_{i_1<i_2}\sum_{k_1=1}^{v}\sum_{k_2=1}^{v}\omega_{i_1,k_1}\omega_{i_2,k_2}f(\xi_{i_1,k_1},\xi_{i_2,k_2}) - (n-2)\sum_{i=1}^{n}\sum_{k=1}^{v}\omega_{i,k}f(\xi_{i,k}) + \frac{(n-1)(n-2)}{2}f(\boldsymbol{\mu})\ . \tag{5.5}$$

Here, for our convenience, we ignore the input variables fixed with their mean values. Expanding the terms $f(\xi_{i_1,k_1},\xi_{i_2,k_2})$ and $f(\xi_{i,k})$ in Taylor's series about the input mean $\boldsymbol{\mu} = (\mu_1,\mu_2,\ldots,\mu_n)$, we get

$$\begin{aligned}\widehat{\mu}_Y &= f(\boldsymbol{\mu}) + \sum_{j=1}^{\infty}\frac{1}{j!}\sum_{i=1}^{n}\frac{\partial^j f(\boldsymbol{\mu})}{\partial\xi_i^j}\sigma_i^j\sum_{k=1}^{v}\omega_{i,k}\alpha_{i,k}^j \\ &+ \sum_{j_1=1}^{\infty}\sum_{j_2=1}^{\infty}\frac{1}{j_1!j_2!}\sum_{i_1<i_2}\frac{\partial^{j_1+j_2}f(\boldsymbol{\mu})}{\partial\xi_{i_1}^{j_1}\partial\xi_{i_2}^{j_2}}\sigma_{i_1}^{j_1}\sigma_{i_2}^{j_2}\sum_{k_1=1}^{v}\omega_{i_1,k_1}\alpha_{i_1,k_1}^{j_1}\sum_{k_2=1}^{v}\omega_{i_2,k_2}\alpha_{i_2,k_2}^{j_2}\ .\end{aligned} \tag{5.6}$$

6 Error computed from MRM for independent system of random variables.

$$\begin{aligned}\text{Error} &= \sum_{j=4}^{\infty}\frac{1}{j!}\sum_{i=1}^{n}\frac{\partial^j f(\boldsymbol{\mu})}{\partial\xi_i^j}\left\{\mathbb{E}(\xi_i-\mu_i)^j - \sigma_i^j\left(\omega_i^+(\alpha_i^+)^j + (-1)^j\omega_i^-(\alpha_i^-)^j\right)\right\} \\ &+ \sum_{j_1=2}^{\infty}\sum_{j_2=2}^{\infty}\frac{1}{j_1!j_2!}\sum_{i_1<i_2}\frac{\partial^{j_1+j_2}f(\boldsymbol{\mu})}{\partial\xi_{i_1}^{j_1}\partial\xi_{i_2}^{j_2}}\,\mathbb{E}(\xi_{i_1}-\mu_{i_1})^{j_1}\mathbb{E}(\xi_{i_2}-\mu_{i_2})^{j_2} \\ &+ \sum_{j_1=2}^{\infty}\cdots\sum_{j_n=2}^{\infty}\frac{1}{j_1!\ldots j_n!}\sum_{i_1<\ldots<i_n}\frac{\partial^{j_1+\cdots+j_n}f(\boldsymbol{\mu})}{\partial\xi_{i_1}^{j_1}\ldots\partial\xi_{i_n}^{j_n}}\,\mathbb{E}(\xi_{i_1}-\mu_{i_1})^{j_1}\ldots\mathbb{E}(\xi_{i_n}-\mu_{i_n})^{j_n}.\end{aligned} \tag{6.1}$$